LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

ANSWER KEY & MARKING SCHEME

Ch 12 Test: Magnetic Effects of Electric Current

Chapter / Topic: Magnetic Effects of Electric Current

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SECTION A

Q1. Which of the following correctly describes the magnetic field near a long, straight, current-carrying wire? **[1]**

Answer: (c) Concentric circles centred on the wire

Q2. The magnetic field inside a long, straight, current-carrying solenoid: **[1]**

Answer: (d) is the same at all points (uniform)

Q3. A positively charged particle (alpha particle) moving towards the west is deflected towards the north by a magnetic field. The direction of the… **[1]**

Answer: (b) upward (by Fleming's left-hand rule)

Q4. Magnetic field lines: **[1]**

Answer: (c) never intersect each other

Q5. Which device protects a household circuit from short-circuiting and overloading? **[1]**

Answer: (d) Fuse (or MCB)

Q6. **Assertion (A):** Two magnetic field lines never cross each other. **Reason (R):** If they crossed, a compass needle placed at the crossing point would have… **[1]**

Answer: (a)

SECTION B

Q7. State Fleming's left-hand rule. **[2]**

Answer: Stretch the thumb, forefinger and middle finger of the left hand so that they are mutually perpendicular. If the forefinger points in the direction of the magnetic field and the middle finger in the direction of the current, then the thumb points in the direction of the force (motion) on the conductor.

- Three fingers mutually perpendicular: 1
- Correct assignment of field, current, force: 1

OR, answer: (Any two) Outside a magnet they emerge from the north pole and merge at the south pole, forming closed curves. They never intersect. Where they are closer together, the field is stronger.

- Any two: 1 + 1

Q8. What is the function of the earth wire? Why should metallic appliances be earthed? **[2]**

Answer: The earth wire is a safety measure that gives a low-resistance path for current. If current leaks to the metal body of an appliance, it flows to the earth, keeping the body at earth potential. This prevents a severe electric shock to the user.

- Low-resistance path to earth: 1
- Prevents shock from leakage: 1

SECTION C

Q9. Answer the following about a solenoid: **[3]**

(i) A coil of many circular turns of insulated copper wire, wrapped closely in the shape of a cylinder. [1]

(ii) The field patterns are similar. One end of the solenoid behaves as a north pole and the other as a south pole. [1]

(iii) Place a soft iron core inside the solenoid and pass a current through it. The core becomes strongly magnetised. [1]

Q10. Answer the following: **[3]**

(i) If you hold a current-carrying straight conductor in your right hand with the thumb pointing in the direction of the current, your fingers curl around the conductor in the direction of the magnetic field lines. [1]

(ii) Directly below: from north to south. Directly above: from south to north. [2]

- Below: 1
- Above: 1

OR, answer: (1) The number of turns in the coil. (2) The strength of the current. (3) The material of the core (a soft iron core makes it much stronger).

- Each factor: 1

SECTION D

Q11. Answer the following about domestic electric circuits: **[5]**

(i) Live wire: red insulation. Neutral wire: black insulation. Earth wire: green insulation. The potential difference between live and neutral is 220 V. [3]

- Three wires with colours: 2
- 220 V: 1

(ii) Short-circuiting: the live and neutral wires come into direct contact (e.g. when insulation is damaged), so the current in the circuit increases abruptly. Overloading: too much current is drawn, e.g. when too many appliances are connected to one socket, or there is an accidental rise in the supply voltage. [2]

- Short-circuit: 1
- Overloading: 1

OR, answer: (a) (i) The force increases. (ii) The direction of the force reverses. (iii) The force becomes zero (the force is greatest when the current is perpendicular to the field). (b) Any two: electric motor, loudspeaker, electric generator, measuring instruments.

- (a) 1 each
- (b) 1 each

SECTION E

Q12. Rohan's house has two separate circuits: one rated 15 A for high-power appliances like the geyser and AC, and one rated 5 A for fans and bulbs. One… **[4]**

(i) High-power appliances draw a large current, so a separate high-rating circuit prevents overloading of the lighting circuit. [1]

(ii) Overloading. The total current drawn was more than the rated value of the circuit. [1]

(iii) $I = P/V = 2000/220 \approx 9.1$ A. This is more than 5 A, so it should not be used on the 5 A circuit. Use the 15 A circuit. [2]

- $I \approx 9.1$ A: 1
- Not safe on 5 A, with reason: 1